

Inductive Coupling and Tuning in NMR Probes; Applications

PHILIP L. KUHNS

Chemagnetics, 208 Commerce Drive, Fort Collins, Colorado 80524

AND

MARTIN J. LIZAK, SAM-HYEON LEE, AND MARK S. CONRADI

Department of Physics, Washington University, St. Louis, Missouri 63130

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Inductive coupling and tuning of NMR probes is ideally suited for certain applications because no direct electrical connections to the resonant circuit are required. The relatively transparent case of series-tuned link coupling is analyzed. It is shown that tuning and coupling adjustments are orthogonal and that most of the RF field comes from the main circuit, not the link. Applications of inductive tuning and coupling are suggested, including low-temperature NMR where Dewar space is restricted. An experimental comparison of conventional, link-coupled, and link-coupled and tuned circuits at 85 MHz is reported. Some novel VHF self-contained resonators are discussed and their performance is experimentally compared to conventional coils. © 1988 Academic Press, Inc.

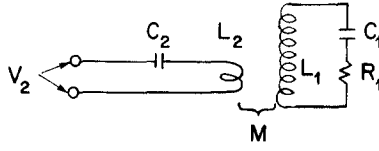
Although inductive coupling and tuning of resonant circuits are old and well-known techniques of radio engineering (1), they are fairly rare in NMR probe (antenna) circuits. It is our opinion that these techniques should be part of the standard repertoire of probe designers. The circuit equations of link-coupled resonant circuits are shown here to be extremely simple, provided the link is series tuned. Unlike most coupling schemes, the coupling and tuning adjustments are orthogonal. In most circumstances, most of the RF magnetic field H_1 comes from the main resonant circuit and not the link.

Inductive tuning and coupling are useful in applications where direct electrical connection to the resonant circuit must be avoided. Such an application involving *in vivo* spectroscopy has already been reported (2). We describe other applications (low-temperature NMR, high-pressure NMR, and VHF resonators) where magnetic coupling and tuning offer distinct advantages. The experimental performance of some novel VHF resonators is reported. Also, inductive coupling and tuning are compared experimentally to conventional methods at 85 MHz.

CIRCUIT ANALYSIS

The circuit of Fig. 1a is a resonant circuit L_1 - C_1 - R_1 inductively coupled to link L_2 . For NMR, the sample is located inside coil L_1 . Because of the resulting simplicities in analyzing and tuning the circuit, link L_2 is series resonated by capacitor C_2 . The

(a)



(b)

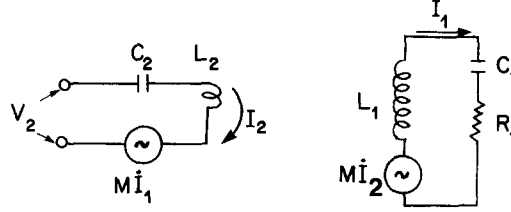


FIG. 1. (a) Series-tuned link, coupled magnetically to tuned circuit L_1 - C_1 . The losses of L_1 are contained in R_1 . (b) The equivalent circuit, with the mutual inductive coupling treated as voltage generators.

mutual inductance M between the two coils can be treated (1) by the equivalent circuit of Fig. 1b. There, voltage generators explicitly show the mutually induced EMFs. Because of $\omega^2 L_1 C_1 = 1$, the total impedance of series circuit 1 is just R_1 :

$$I_1 = \frac{1}{R_1} M \dot{I}_2. \quad [1]$$

The terminal voltage V_2 of the link circuit is just the induced voltage $M \dot{I}_1$, because L_2 and C_2 are series resonant at the operating frequency ω . Using Eq. [1], V_2 is

$$V_2 = M \dot{I}_1 = \frac{M^2}{R_1} \ddot{I}_2. \quad [2]$$

The term $\ddot{I}_2$ is in-phase with I_2 and can be written as $\omega^2 I_2$.

The mutual inductance M is conveniently expressed (1) in terms of the (dimensionless) coefficient of coupling K :

$$K^2 = \frac{M^2}{L_1 L_2}. \quad [3]$$

The value of K can vary from zero (no coupling) to unity (complete sharing of flux by the coils). Equation [2] becomes

$$V_2 = \frac{M^2 \omega^2 L_1 L_2 I_2}{L_1 L_2 R_1} = K^2 \left(\frac{\omega L_1}{R_1} \right) (\omega L_2) I_2. \quad [4]$$

Thus, the impedance R_{in} of circuit 2 seen from its terminals is resistive and is just

$$R_{in} = \frac{V_2}{I_2} = K^2 Q_1 (\omega L_2), \quad [5]$$

where Q_1 is the unloaded Q of circuit 1.

Typical values in Eq. [5] for NMR are $Q_1 = 100$ and $\omega L_2 = 50 \Omega$, corresponding typically to a two-turn link L_2 . The required coupling coefficient for an input resistance $R_{in} = 50 \Omega$ is $K = 0.1$, an easily obtained degree of coupling. In this example, the loaded Q of the link circuit is one, demonstrating that exact tuning of the link is not important. In practice, we select a fixed capacitor for C_2 that tunes L_2-C_2 to within 10% of the operating frequency (in the absence of L_1-C_1).

The tuning of the circuit of Fig. 1 is accomplished by adjustment of C_1 or L_1 . The coupling adjustment is made by physically moving link L_2 toward or away from L_1 . The orthogonality of the tuning and coupling adjustments is indicated in the separate and uncoupled equations $\omega^2 L_1 C_1 = 1$ and Eq. [5]. This orthogonality is the result of series tuning the link and makes probe tuning easy and fast. The loaded Q of the coupled circuits may be reduced by overcoupling, without changing the resonance frequency. The equations governing untuned links (3) are rather complicated, by comparison. Another analysis of inductive coupling has appeared, but the approach was to use a separate matching circuit between the link and the 50Ω coaxial cable (2).

Another benefit of tuning the link is that it reduces the coupling coefficient required. With an untuned link, the circuit 1 must be detuned slightly to cancel the self-inductance L_2 . Below some critical coefficient of coupling, the input impedance of the link cannot be made real (for example, see Fig. 3 of Ref. (3)).

What are the best values of link inductance L_2 and coupling K to use in Eq. [5]? In general the coupling should be made as large as possible. This results in the smallest link inductance L_2 and thus the lowest loaded Q (and least sensitive tuning) of the link circuit.

An interesting question for NMR with link coupling is what fractions of the RF field come from L_1 and from L_2 . The magnetic stored energy can be written as (4)

$$U = U_1 + U_2 + U_{12} = \frac{1}{2} L_1 I_1^2 + \frac{1}{2} L_2 I_2^2 + M I_1 I_2. \quad [6]$$

Using Eqs. [1] and [3], the ratio of stored energy in coils 1 and 2 can be expressed as

$$\frac{U_1}{U_2} = \frac{\frac{1}{2} L_1 I_1^2}{\frac{1}{2} L_2 I_2^2} = \frac{L_1}{L_2} \left(\frac{M \omega}{R_1} \right)^2 = K^2 Q_1^2. \quad [7]$$

For the typical values mentioned above $K^2 Q_1 \simeq 1$, so $U_1 \gg U_2$ provided $Q_1 \gg 1$, as is common. More generally, using the result of Eq. [5],

$$\frac{U_1}{U_2} = \left(\frac{R_{in}}{\omega L_2} \right) Q_1. \quad [8]$$

This shows the advantages of using a large coupling K and small link inductance L_2 : keeping most of the stored energy in resonant circuit 1.

The energy in the cross-term U_{12} is usually small. Using Eq. [1], the ratio U_1/U_{12} is

$$\frac{U_1}{U_{12}} = \frac{\frac{1}{2} L_1 I_1^2}{M I_1 I_2} = \frac{L_1 \omega}{2 R_1} = \frac{Q_1}{2}. \quad [9]$$

Provided $Q_1 \gg 1$, the cross-term magnetic energy is negligible. The RF field seen by the sample residing in coil 1 will arise almost entirely from current in coil 1, in the typical application.

APPLICATIONS

Use of inductive tuning and coupling for *in vivo* spectroscopy has been reported (2). Wires passing into the subject animal's body have been eliminated while retaining the high sensitivity and spacial selectivity of a resonant circuit wrapped around the desired organ.

In low-temperature NMR experiments, there is often very little working space. The NMR circuitry must compete for space with heat exchangers, Dewar walls, and refrigerators. The use of conventional, high-voltage tuning capacitors may simply not be possible because of lack of space, differential contraction and freezing in the capacitor, and RF breakdown due to gases. We propose the arrangement of Fig. 2, suitable for iron-core magnets. Sample coil L_1 is tuned by fixed capacitor C_1 , chosen to resonate L_1 slightly below the operating frequency. The shorted turn T below L_1 is an inductive tuner. It can be moved closer to L_1 , reducing the inductance of L_1 and raising the resonant frequency. Coupling to the resonant structure is obtained with a series-tuned link. This axial geometry makes good use of the space in most Dewars. The tuning and coupling adjustments do not involve the handling of high RF voltages. Even the voltage on capacitor C_1 may be reduced by using a low L_1/C_1 ratio; because C_1 can be located immediately at L_1 , negligible magnetic energy will be associated with the current in C_1 's leads. The major disadvantage to the design of Fig. 2 is that the tuning range is limited ($\sim 5\%$).

In high-pressure NMR, it is generally required to pass the RF coil wire through the pressure wall by means of a high-pressure feedthrough. Even the best feedthroughs have limited temperature ranges and limited reliability. We propose that magnetic coupling and tuning be accomplished by use of a high-pressure optical window. Coils

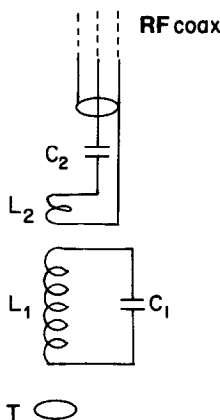


FIG. 2. Proposed design of RF circuitry in low-temperature NMR. The location of shorted tuning loop T is adjusted to tune sample coil L_1 . Coupling is adjusted by the location of series-tuned link L_2 . The axial design makes efficient use of the limited low-temperature working space.

would be placed on opposite sides of the window to obtain coupling from the atmospheric pressure region into the high-pressure vessel.

EXPERIMENTAL COMPARISON

We constructed three probe circuits for 85 MHz proton NMR. The circuits, shown in Fig. 3, include a conventional capacitively tuned and coupled device along with an inductively coupled circuit and an inductively tuned and coupled version. The coils L_1 in Figs. 3a and 3b are six turns of 18 AWG closewound on a 5 mm diameter form. The links L_2 are two turns of 20 AWG hook-up wire; C_2 is 47 pf. In Figs. 3a and 3b the tuning capacitors are ceramic trimmers. The NMR samples were glycerol in 1.5 mm i.d. tubes; the samples extended well above and below the RF coils.

The coil L_1 in Fig. 3c is 0.1 mm thick copper sheet about 7 mm wide wound on a 5 mm diameter form. It is resonated by a total of 1400 pf in the form of nonmagnetic, porcelain chip capacitors. The inductance of the coil is only 2.5 nH. The Q of this circuit is roughly 1/2 that of Figs. 3a and 3b; we believe this to be due to the resistance of the soft solder used to connect the chip capacitor in the particularly low L/C circuit. This resonator is a highly loaded version of the loop-gap resonator (5, 6).

As expected, the link-coupled circuits show orthogonal tuning and coupling adjustments. The correct locations of the link and shorted turn T in Fig. 3c are not

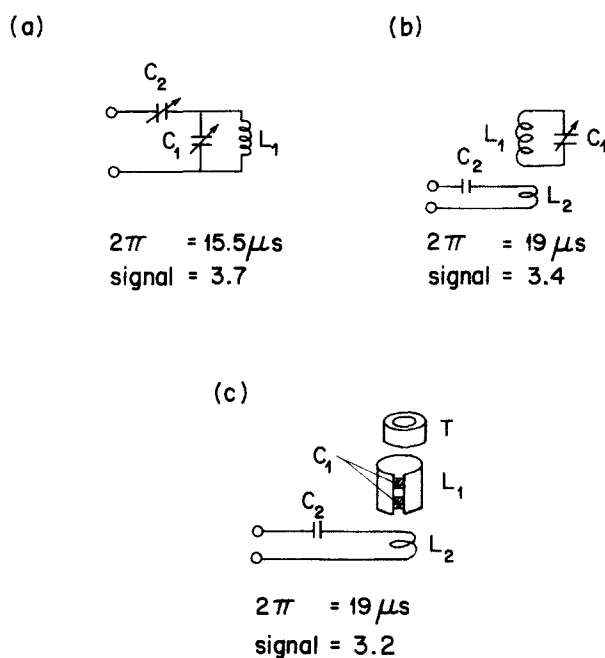


FIG. 3. Three probe circuits constructed for experimental comparison. (a) Conventional capacitive coupling and tuning; (b) link coupling and conventional tuning; (c) link coupling with inductive tuning, using particularly low L/C ratio. The pulse lengths for 2π nutation and the relative signal amplitudes after a $\pi/2$ pulse are shown below each circuit.

critical; mechanical precision is not needed. We fixed the locations with glue. The low L/C resonator of Fig. 3c is insensitive to dielectric effects: not only does the sample not change the tuning, we are able to place our fingers all over the structure with essentially no detuning or change in Q .

In Fig. 3 we report the relative signal strengths after $\pi/2$ pulses of the three circuits. Also reported are the times required for the 25 W transmitter to flip the spins through 2π . Clearly, the performances are comparable, though the conventional circuit has slightly better performance.

LOW L/C RESONATORS FOR VHF

One distinct advantage of inductive coupling and tuning is that it frees the probe designer to consider unusually low L/C ratios in the resonant circuit. Generally, low L/C structures have very low voltages and are not subject to RF breakdown, even at high powers. Further, the electric field in the sample is small with low L/C circuits, keeping sample heating and detuning by the sample at a minimum. This is particularly important for biological samples (7). But it is difficult to tune such circuits by conventional variable capacitors: the inductance of the connecting wires will typically exceed that of the coil.

Therefore, we propose the self-contained resonators of Figs. 4 (solenoidal) and 5 (saddle-coil style). The capacitance necessary to resonate the one-turn coil is formed from the overlapping conductors. The design of Fig. 4 is correctly regarded as a variation of the split-ring (5) or loop-gap resonator (6). The Fig. 5 design may be viewed as a distributed capacitance version of the design by Alderman and Grant (7). In this context it is interesting to note that the first NMR was done over 40 years ago in a very low L/C 30 MHz cavity filled with paraffin wax (8). It should be noted that slotted (9, 10) and folded (11) $\lambda/2$ structures have been used for NMR and ESR, as well as a sectionalized coil (12) with low electric fields in the sample.

There are several ways to implement the proposed VHF resonators. One can use vapor deposition of aluminum or copper, electro-plating, or electro-less deposition. There are various aluminized Mylar films available, as well as aluminum duct tape with one side adhesive. In addition, very thin Teflon-copper laminates are available

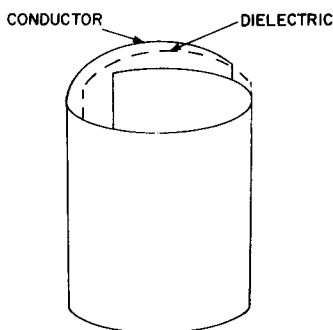


FIG. 4. Self-contained resonator of solenoidal design. This is essentially a wide, one-turn inductor resonated by the capacitance between the overlapping conductors. The dielectric is Teflon tape and is represented by the dashed curve.

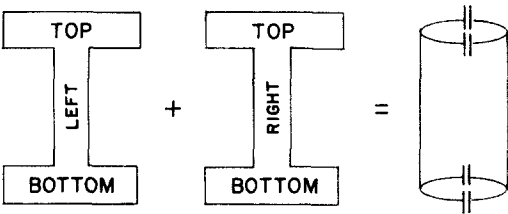


FIG. 5. Self-contained resonator of saddle-coil design, formed from two identical pieces of copper (shown flat, before wrapping onto glass tube with axis running from top to bottom). The length of the top and bottom tabs is slightly less than the tube circumference. The completed resonator is drawn schematically, with each capacitor representing a region of overlapping of the two conductors with Teflon tape dielectric.

as microwave circuit board. However, at 340 MHz (the proton frequency in our high-field magnet), large capacitances are needed to resonate single-turn coils of ~ 10 mm diameter. This ruled out many of the above schemes.

We finally settled on a resonator of plain copper or aluminum foil, with a Teflon pipe tape dielectric in the overlap region (see Fig. 4). At 340 MHz, the dielectric must be chosen carefully. The resonant frequency of the device is roughly adjusted by trimming the amount of overlap of the conductors. The resonator is then wrapped with ordinary cellophane tape, to keep the circuit from unwinding. The resonators as built were rugged enough to allow proof-of-principle, but certainly would be unsatisfactory for routine use. Link coupling and a copper shorted turn (inductive fine tuning) were used on opposite ends of the resonator, as in Fig. 3c. In Table 1 we list the times for a π pulse with our ~ 20 W transmitter, for copper and aluminum resonators and for two- and four-turn ordinary coils of wire. The results for a two-turn coil of flat ribbon are also indicated. All circuits were link coupled. The coils of wire and ribbon were tuned by small ceramic trimmer capacitors. All of the devices had approximately the same active volumes and shapes. The relative signal amplitudes for a standard glycerol sample are shown in Table 1. Also listed are the approximate loaded Q ($=\frac{1}{2}$ unloaded Q) and amount of frequency detuning by the standard sample. Because of variations in sample location and exact active volume, the pulse lengths for π nutations are the most reliable measure of circuit performance.

The results in Table 1 indicate that the self-contained resonators are matched in performance (H_1 and signal amplitude) only by the two-turn ribbon. The resonators show very little detuning by the dielectric effect of the sample, as expected.

TABLE 1
Performance Comparison of 340 MHz Circuits

Circuit	π Pulse (μ s)	Signal amplitude	Loaded Q	Sample detuning (MHz)
Aluminum resonator	26	121	150	1
Copper resonator	23	88	150	1
Two turns wire	95	—	70	1.5
Four turns wire	36	61	70	7
Two turns ribbon	24	88	120	2
Saddle resonator	38	84	120	1

We also constructed a self-contained resonator of the saddle-coil type, for vertical loading of sample tubes. We started with the saddle coil shown in Fukushima and Roeder's book (13). Again, copper foil and Teflon tape were used with cellophane tape securing all. The series-tuned link and copper loop tuner were placed on opposite sides of the coil. The construction of the saddle resonator is shown in Fig. 5. The active volume of this device is similar to the volume of the solenoidal circuits compared in Table 1. As listed in Table 1, the H_1 of the saddle resonator is about a factor of two weaker than our best solenoidal designs; this is approximately what is found with ordinary coils at lower frequencies.

It should be remarked that our goal was to demonstrate that the resonators are competitive with conventional circuits; we made no efforts to optimize any devices.

CONCLUSIONS

The analysis and tuning of link-coupled probe circuits are simplified by series resonating the link. The required coupling coefficient K is also reduced with a tuned link. The experimental measurements demonstrate that the performance of inductively coupled and/or inductively tuned probe circuits is comparable with that of conventional designs.

Inductive coupling and tuning will be useful when it is difficult to connect directly to the resonant circuit—as with *in vivo* spectroscopy and high-pressure NMR. In low-temperature NMR, inductive tuning and coupling eliminate the need for high-voltage tuning capacitors in the limited cryostat volume. Finally, we have proposed self-contained resonators for VHF NMR. These resonators have low voltages (avoiding RF breakdown in high-power NMR), low electric fields (minimizing detuning by the sample), and high Q . Experimentally, the self-contained resonators perform as well as or better than coils of ribbon or wire.

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